LECTURE 2:
CONSTRUCTION OF THE LEBESGUE MEASURE

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to elre@princeton.edu, thanks in advance!

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How do we define the uniform measure on $[0,1]$**?** Recall that the notation $[a,b]$ denotes the closed interval between a and b (i.e., including the endpoints a and b), and (a,b) denotes the open interval (i.e., excluding a and b).

- $\mathbb{P}(\{a\}) = 0$ for any $a \in [0,1]$. (Why?)
- $\mathbb{P}([0,1]) = 1$, and $\mathbb{P}[0,\frac{1}{2}] = \mathbb{P}[\frac{1}{2},1] = \frac{1}{2}$. Basically, we know how we should define the probability of any segment:
$$\mathbb{P}([a,b]) = \mathbb{P}((a,b)) = \mathbb{P}((a,b]) = |b-a|,$$
i.e., by its length.
- What about other sets? Disjoint (countable) unions of the segments have probabilities equal to their joint length (by countable additivity). Think through what should happen with the intersections of the segments. What about open intervals? What about the measure of the Cantor set?

Note that we did not define a σ-algebra on $[0,1]$ yet; however, we already see that we need to think about *sets*, and not just about individual outcomes.

Lebesgue measure construction

Let's formally construct the uniform (probability) measure on $(0,1]$ equipped with the Borel σ-algebra $\mathcal{B}((0,1])$, which is called the **Lebesgue measure**. We consider the sample space $(0,1]$ for convenience. (To construct the Lebesgue measure on $[0,1]$, we can simply add a point 0 such that $\{0\}$ has measure zero without complications.)

First, observe that the collection

$$\mathcal{L} := \left(\bigcup_{i=1}^{n} (a_i, b_i] : 0 \le a_1 < b_1 < a_2 < b_2 < \ldots < a_n < b_n \le 1;\ n \text{ is finite, can be } 0 \right) \tag{1}$$

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is an algebra of sets (note that we take $\emptyset \in \mathcal{L}$), and that $\sigma(\mathcal{L}) = \mathcal{B}((0,1])$. Define the function $\lambda : \mathcal{L} \to [0,1]$ by

$$\lambda\left(\bigcup_{i=1}^{n}(a_i, b_i] : 0 \leq a_1 < b_1 < a_2 < b_2 < \ldots < a_n < b_n \leq 1,\ 0 \leq n < \infty\right) := \sum_{i=1}^{n}(b_i - a_i). \quad (2)$$

This defines a function that assigns a measure to finite disjoint unions of semi-intervals in an intuitive way (i.e., as their total length). In the following, we will extend the definition of λ to the Borel σ-algebra $\mathcal{B}((0,1])$ on $(0,1]$ generated by $\mathcal{L}$.

Definition 2.1. A function $\mu : 2^{\Omega} \to [0,\infty]$ supported on an algebra $\mathcal{A}$ is called a **measure on a set algebra** if

(1) $\mu(\emptyset) = 0$.
(2) $\mu(A) \geq 0$ for any $A \in \mathcal{A}$.
(3) μ is a countably additive function, namely, for any countable disjoint collection of subsets $A_i \in \mathcal{A}$, such that $\sqcup_i A_i \in \mathcal{A}$, $\mu(\sqcup_i A_i) = \sum_i \mu(A_i)$.

Note the similarity with the definition of a measure on a σ-algebra from Lecture 1; the only difference here is that we only need the countable additivity property to hold for countable disjoint unions that also belong to the algebra.

Proposition 2.2. *The function λ is a measure on $\mathcal{L}$.*

Theorem 2.3 (Carathéodory extension theorem)**.** *Let $\mathcal{A}$ be an algebra of subsets of Ω. If $\mu : \mathcal{A} \to [0,\infty]$ is a measure on $(\Omega, \mathcal{A})$, then μ has an extension on $\sigma(\mathcal{A})$: there exists a measure $\bar{\mu} : \sigma(\mathcal{A}) \to [0,\infty]$ on $(\Omega, \sigma(\mathcal{A}))$ such that $\mu(A) = \bar{\mu}(A)$ for all $A \in \mathcal{A}$. If μ is a finite measure ($\mu(\Omega) < \infty$), then the extension $\bar{\mu}$ is unique.*

First, think through how Proposition 2.2 and Theorem 2.3 together imply the existence and uniqueness of the extension of λ to the Borel σ-algebra $\mathcal{B}((0,1])$ on $[0,1]$. This (unique) extension is called the *Lebesgue measure* on $(0,1]$.

Remark 2.4. The uniqueness of the extension from Carathéodory's theorem (Theorem 2.3) actually holds in the case where μ is σ-finite (see the remark in Lecture 1). This allows us to prove the existence and uniqueness of the Lebesgue measure on the real line.

Proofs

We will discuss the proofs of Proposition 2.2 and Theorem 2.3. Note that non-negativity and finiteness of λ is obvious (since the length of the interval $[0,1]$ is bounded). Hence, to prove Proposition 2.2, it suffices to show that λ is countably additive on $\mathcal{A}$.

2.1. Countable additivity of λ.

Proof of Proposition 2.2. First, note that $\lambda((0,1]) = 1$, $\lambda(\emptyset) = 0$, and that λ is a non-negative monotone function. It is also easy to see that λ satisfies finite additivity (since the union of a finite set of semi-intervals simply consists of longer or more disjoint semi-intervals). In particular, for any finite collection of sets $L_1, \ldots, L_k \in \mathcal{L}$, the following *subadditivity property* holds:

$$\lambda\left(\bigcup_{i=1}^{k} L_i\right) \leq \sum_{i=1}^{k} \lambda(L_i). \quad (3)$$

Now, the whole point is to check countable additivity. It is enough to check that

$$\lambda(L) = \sum_{i=1}^{\infty} \lambda(L_i) \quad \text{for any disjoint } L_i \text{ and } L = \sqcup_{i=1}^{\infty} L_i \text{ so that } L_i, L \in \mathcal{L}.$$

One side is easy: for any $k < \infty$,

$$\lambda(L) \geq \lambda\left(\bigsqcup_{i=1}^{k} L_i\right) = \sum_{i=1}^{k} \lambda(L_i).$$

Hence, by taking $k \to \infty$, we deduce that

$$\lambda(L) \geq \sum_{i=1}^{\infty} \lambda(L_i). \tag{4}$$

For the reverse inequality, our goal is to approximate L by a *finite union* of the sets L_i that is not much smaller than L, and use (3). The key idea is to use the fact that for any compact set, any open cover has a finite subcover.[1] So, we will create a collection of closed (and hence compact) sets $L'(\delta)$ that are slightly smaller approximations of L, and a collection of open sets $L_i'(\varepsilon)$ for each i that are slightly larger approximations than L_i.

Since L and each L_i are in $\mathcal{L}$, we can write

$$L = \bigsqcup_{j=1}^{\ell} (a_j, b_j] = \bigsqcup_{i=1}^{\infty} \bigsqcup_{j=1}^{\ell_i} (a_{ij}, b_{ij}], \quad \text{and} \quad L_i = \bigsqcup_{j=1}^{\ell_i} (a_{ij}, b_{ij}]$$

for some finite sets of endpoints a_j, b_j (for L), and a_{ij}, b_{ij} (for each L_i). For any $\delta > 0$, let

$$L'(\delta) := \bigsqcup_{j=1}^{\ell} \left[a_j + \frac{\delta}{2^j}, b_j\right], \quad \text{and} \quad L_i'(\varepsilon) := \bigsqcup_{j=1}^{\ell_i} \left(a_{ij}, b_{ij} + \frac{\varepsilon}{2^{i+j}}\right).$$

Note that $L'(\delta)$ is compact, $L'(\delta) \subset L$, and $\lambda(L) \leq \lambda(L'(\delta)) + \delta$.[2] Furthermore, each $L_i'(\varepsilon)$ is an open set, $L_i \subset L_i'(\varepsilon)$, and $\lambda(L_i'(\varepsilon)) \leq \lambda(L_i) + \varepsilon 2^{-i}$. Since $L = \sqcup_{i=1}^{\infty} L_i$, from this construction, we have

$$L'(\delta) \subseteq \bigcup_{t=1}^{k} L_{i_t}'(\varepsilon) \subseteq \bigcup_{i=1}^{\infty} L_i'(\varepsilon)$$

for some finite set of indices $i_1, \ldots, i_k$, since any open cover of a compact set always has a finite subcover. Thus, we deduce that

$$\lambda(L) - \delta \leq \lambda\left(\bigcup_{t=1}^{k} L_{i_t}'(\varepsilon)\right) \overset{(3)}{\leq} \sum_{t=1}^{k} \lambda(L_{i_t}'(\varepsilon)) \leq \sum_{i=1}^{\infty} \lambda(L_i'(\varepsilon)) \leq \sum_{i=1}^{\infty} \lambda(L_i) + \varepsilon.$$

By taking ε and δ as small as we like, we deduce that $\lambda(L) \leq \sum_{i=1}^{\infty} \lambda(L_i)$, which is the reverse inequality to (4). Hence, we conclude that λ is countably additive on $\mathcal{L}$. $\square$

[1] This is the *Heine-Borel theorem*: finite unions of closed segments on $\mathbb{R}$ are compact (i.e., closed and bounded).

[2] Note that technically $L'(\delta)$ and $L_i'(\varepsilon)$ are not in $\mathcal{L}$. But each of these sets differs from a measurable set in $\mathcal{L}$ by finitely many points: i.e., we may obtain the sets by adding or subtracting the endpoints of the intervals from sets in $\mathcal{L}$. Thus, we may simply consider adding all singletons (i.e., sets of single points) to $\mathcal{L}$ and defining their measure to be zero according to λ.

Now let's discuss the ideas behind the Carathéodory extension theorem (Theorem 2.3).

2.2. **Carathéodory extension theorem proof ideas: existence.** The existence of the extension is based on defining the following so-called *outer measure* based on μ: for $A \in 2^\Omega$,

$$\mu^*(A) := \inf \left\{ \sum_i \mu(A_i) : \ A_1, A_2, \ldots \in \mathcal{A} \text{ and } A \subseteq \cup_i A_i \right\}. \tag{5}$$

Intuitively, we approximate the set A in question by a covering union ("outer approximation") of the events that we can measure by μ, and use this to *define* the measure of A by the value of the tightest cover like this.

The pieces of the formal proof are as follows. In general, an outer measure is defined in the following way: it is any set function $\phi : 2^\Omega \to [0, \infty]$ such that

- $\phi(\emptyset) = 0$.
- $\phi(A) \leq \phi(B)$ for all $A \subseteq B \subseteq \Omega$.
- $\phi(\bigcup_{i=1}^\infty A_i) \leq \sum_{i=1}^\infty \phi(A_i)$ for all $A_i \subseteq \Omega$.

Interestingly, the outer measure ϕ is not defined on a σ-algebra, but rather defines a σ-algebra by itself (a set A is measurable if $\phi(B) = \phi(B \cap A) + \phi(B \cap A^c)$ for any B).

It can be checked that μ^*, defined in (5), is indeed an outer measure based on μ; that all the sets in $\mathcal{A}$ are μ^*-measurable; and that $\mu^*(A) = \mu(A)$ for all $A \in \mathcal{A}$. Therefore, the σ-algebra generated by the outer measure μ^* contains $\sigma(\mathcal{A})$, and μ^* can serve as the desired extension of μ to $\sigma(A)$.

2.3. **Carathéodory extension theorem proof ideas: uniqueness.** The uniqueness of the extension follows from the following theorem. Before stating the result, we introduce some additional definitions.

- Any collection $\mathcal{P}$ of subsets of Ω that is closed under finite intersections is called a **π-system**.
- Any collection Λ of subsets of Ω that (a) contains Ω, (b) contains set complements (if A, B are in the system and $A \subset B$ then $B \setminus A$ is in the system), and (c) is closed under monotone increasing limits (if $A_1 \subseteq A_2 \subseteq \ldots$ are all in the system then $\cup_i A_i$ is also in the system) is called a **λ-system**.

Theorem 2.5 (Dynkin[3] π-λ theorem)**.** *If $\mathcal{P}$ is a π-system on Ω, and $\mathcal{P} \subseteq \Lambda$, where Λ is a λ-system on Ω, then $\Lambda \supseteq \sigma(\mathcal{P})$.*

Remark 2.6. Observe that any algebra of sets is a π-system (but not necessarily a λ-system). Actually, a collection of subsets is a σ-algebra if and only if it is a π-system and a λ-system (exercise!).

Proof of Dynkin's π-λ theorem. Let's assume that Λ is the smallest λ-system that contains $\mathcal{P}$. This can be done without the loss of generality after a check (exercise!) that an intersection of λ-systmes must be a λ-system as well.

The key idea of the proof is to conclude that the λ-system Λ is also a π-system. Then, by Remark 2.6 above, Λ must be also a σ-algebra and thus contain $\sigma(\mathcal{P})$, which is the smallest σ-algebra that includes $\mathcal{P}$.

[3]Eugene Dynkin is a mathematician and one of the advisors of Prof. Vanderbei from ORFE!

Let's check that Λ is a π-system. To do this, fix some $A \in \mathcal{P}$ and consider

$$\mathcal{G}_A := \{B \in \Lambda : A \cap B \in \Lambda\}.$$

Then, $\mathcal{G}_A$ contains $\mathcal{P}$ since $\mathcal{P}$ is closed under the intersection and is a subset of Λ. Also, $\mathcal{G}_A \subseteq \Lambda$. One can check by definition and set theory that $\mathcal{G}_A$ is a λ-system itself. So, $\mathcal{G}_A = \Lambda$ by the minimality (the first paragraph of the proof). This is helpful because now we know that if $A \in \Lambda$ and $B \in \Lambda = \mathcal{G}_A$, then$A \cap B \in \Lambda$ (this is by definition of $\mathcal{G}_A$!) This proves that Λ is closed under finite intersections (by induction for more than two sets) and so Λ is also a π system and the proof is done: Λ has to be a σ-algebra and contain $\sigma(\mathcal{P})$. $\square$

The following corollary will be used several times.

Proposition 2.7. *Let $\mathcal{P}$ be a π-system. If two measures μ_1, μ_2 on $(\Omega, \sigma(\mathcal{P}))$ agree on $\mathcal{P}$, and also $\mu_1(\Omega) = \mu_2(\Omega) < \infty$, then $\mu_1 = \mu_2$ on $\sigma(\mathcal{P})$.*

Proof. Let

$$\Lambda := \{A \in \sigma(\mathcal{P}) : \mu_1(A) = \mu_2(A)\}.$$

Then $\mathcal{P} \subseteq \Lambda$. Let's check that Λ is a λ-system. Indeed, $\Omega \in \Lambda$ by assumption. For any $A, B \in \Lambda$, by finite additivity of the measures,

$$\mu_1(B \setminus A) = \mu_1(B) - \mu_1(A) = \mu_2(B) - \mu_2(A) = \mu_2(B \setminus A).$$

(Since we assume finiteness of the measure, no ambiguity can appear as a result of subtracting infinities. See [Chatterjee, Theorem 1.3.6] for how a similar result can be formulated in the σ-finite case.) The last property of a λ-system follows from the continuity from below of measures: if $A_1 \subseteq A_2 \subseteq \dots$ are all in Λ and $A = \cup_i A_i$ then

$$\mu_1(A) = \lim_i \mu_1(A_i) = \lim_i \mu_2(A_i) = \mu_2(A).$$

Hence, we conclude by Dynkin's π-λ theorem (Theorem 2.5). $\square$

We can now prove the uniqueness of the extension for finite measures from Carathéodory's theorem.

Proof of Theorem 2.3: uniqueness. Let μ^* and μ^{**} be any two extensions of μ that are defined on $\sigma(\mathcal{A})$ and agree on the algebra $\mathcal{A}$. Any algebra is a π-system that contains Ω, so, $\mu^*(\Omega) = \mu^{**}(\Omega)$. Hence, by Proposition 2.7 above, μ^* and μ^{**} must agree on the whole generated σ-algebra $\sigma(\mathcal{A})$. $\square$

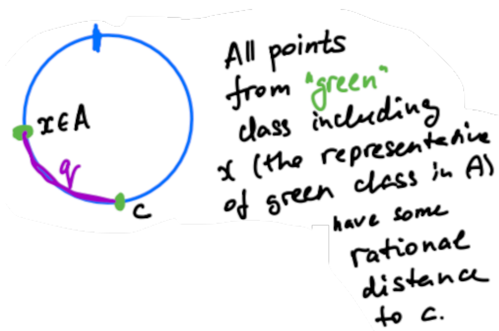
Non-measurable sets

If the state space is uncountable (for example, $\Omega = [0, 1]$ or $\Omega = \mathbb{R}$), we cannot assign probabilities to individual outcomes, and the concept of a σ-algebra is needed to be able to define a "meaningful" measure (i.e., we cannot always take $\mathcal{F} = 2^\Omega$).

Remark 2.8 (Construction of a non-measurable set)**.** To make our life simpler, let's discuss an example of a non-measurable set for the uniform measure on $\mathcal{C}$, a circle of perimeter 1 (which we can identify with the points in $(0, 1]$). Suppose that we have the uniform measure μ on $(\mathcal{C}, \mathcal{F})$ for some σ-algebra $\mathcal{F} \subseteq 2^{\mathcal{C}}$, such that $\mu(\mathcal{C}) = 1$ and $\mu([a, b]) =$ length from a to b. A natural feature of the uniform measure that we require is *shift-invariance*: i.e., $\mu(A + x) = \mu(A)$ for any rotation $x \in \mathcal{C}$ and $A \in \mathcal{F}$.

High-level idea: it is possible to construct a set A such that all of its rational rotations cover the entire circle $\mathcal{C}$, but it does not intersect with any of its rational rotations. Thus, if we have $\mu(A) = 0$, then $\mu(\mathcal{C}) = 0$ by σ-additivity, which is wrong. But if $\mu(A) = a > 0$ then $\mu(\mathcal{C}) = \sum_{i=1}^{\infty} a = \infty$ by σ-additivity, which is also wrong. Hence, we cannot assign a measure to A in any non-contradictory way.

Here is how we can construct a set A as described above. For $a, b \in \mathcal{C}$, we will define the equivalence relation $a \sim b$ if the length of a to b is *rational*. All the points in $\mathcal{C}$ are split into equivalence classes induced by the relation $\sim$. Now, choose one point from each equivalence class (using the axiom of choice) and form the set $A \in 2^{\mathcal{C}}$. What can the measure of A be according to μ?



Let $q_1, q_2, \ldots$ be an enumeration of all the rational numbers in $\mathcal{C}$. We claim that $\sqcup_i (A + q_i) = \mathcal{C}$. To see this, first note that the union is disjoint since we cannot have two points from the same equivalence class in A: if $x \in A + q_1$ and $x \in A + q_2$, then $a_1 + q_1 = a_2 + q_2$ for some $a_1, a_2 \in A$, and hence $a_1 - a_2 \in \mathbb{Q}$, so they are two elements from the same equivalence class. Furthermore, the union covers the whole circle since for any $c \in \mathcal{C}$, there is a point $x \in A$ from the same equivalence class as c and a rotation q such that $c = x + q$.

Think about how to augment this example to show that we cannot have $\mathcal{F} = 2^{\Omega}$ for the uniform probability measure on $[0, 1]$ (for more information, see the construction of the Vitali set).